WORKED SOLUTION

STEP II 1993, Question 1

Pure • Combinatorial Game Strategy

> **STEP II 1993, Question 1**
>
> Read the question before you go on. It is in the STEP Support Programme's question database:
>
> step.maths.org/questions/93-s2-q1

The key idea

Everything rests on one counting fact. Both players have exactly 9 to share out, so Betty cannot match or beat Adam on every field and win outright on one: that would cost her at least $9+1$. So whenever Betty wins, she must win exactly two fields and lose the third. Once you have this, the first part becomes a recipe, and the two-triple part becomes a short case analysis.

Solution

Throughout, "Betty wins field i" means $b_i > a_i$, "Adam wins field i" means $a_i > b_i$, and a field with $a_i = b_i$ is tied.

A lemma: how Betty can win

Suppose Betty's triple beats Adam's triple (a_1, a_2, a_3), where $a_1 + a_2 + a_3 = 9$.

- She cannot win all three fields, since that would need $b_i \geq a_i + 1$ for each i, giving $b_1 + b_2 + b_3 \geq 12$.
- Suppose she wins exactly one field. Since she has more points than Adam, Adam scores 0, so the other two fields are ties or Betty wins. Then $b_i \geq a_i$ for every i, with strict inequality at least once, and so $b_1 + b_2 + b_3 \geq 9 + 1 = 10$. That's impossible.

She has more points than Adam, so she scores at least one. Hence she wins *exactly two* fields. So Betty beats (a_1, a_2, a_3) if and only if, for some pair of fields i, j,

$$b_i \geq a_i + 1, \qquad b_j \geq a_j + 1,$$

and she loses the remaining field k. Note that losing field k is automatic, since $b_k = 9 - b_i - b_j \leq 9 - a_i - a_j - 2 = a_k - 2$.

Betty can always win

Label the fields so that $a_1 \geq a_2 \geq a_3$. Then $3a_1 \geq a_1 + a_2 + a_3 = 9$, so $a_1 \geq 3$. Betty gives up the field where Adam is strongest and beats him by one on the other two:

$$(b_1, b_2, b_3) = (a_1 - 2,\ a_2 + 1,\ a_3 + 1).$$

These are non-negative integers (as $a_1 - 2 \geq 1$) with sum 9. Betty wins fields 2 and 3 and loses field 1, because $a_1 - 2 < a_1$. She scores 2 to Adam's 1, so she wins.

Adam can make sure he never wins

Take $(a_1, a_2, a_3) = (9, 0, 0)$. Adam cannot score on fields 2 or 3, because $b_2, b_3 \geq 0$. So he scores at most one point, and only by winning field 1, which means $b_1 \leq 8$. In that case $b_2 + b_3 = 9 - b_1 \geq 1$, so Betty has a positive entry on field 2 or field 3 and scores at least one point. Either Adam scores 0, or he scores 1 and Betty scores at least 1. In neither case does he score more than Betty, so he never wins, whatever Betty does.

(A brute-force search shows that $(9, 0, 0)$ and its rearrangements are the *only* triples with this property. Against any other triple, some foolish choice by Betty hands Adam the game.)

Two triples: $(5, 3, 1)$ and $(3, 1, 5)$

We need to show that no triple (b_1, b_2, b_3) beats both. By the lemma, Betty needs to win two fields against each triple.

Against $(5, 3, 1)$. Winning fields 1 and 2 costs $6 + 4 = 10 > 9$, which is impossible. So she must win field 3 together with field 1 or field 2:

$$b_3 \geq 2 \quad \text{and} \quad (b_1 \geq 6 \ \text{ or } \ b_2 \geq 4). \tag{A}$$

Against $(3, 1, 5)$. Winning fields 1 and 3 costs $4 + 6 = 10 > 9$, which is impossible. So she must win field 2 together with field 1 or field 3:

$$b_2 \geq 2 \quad \text{and} \quad (b_1 \geq 4 \ \text{ or } \ b_3 \geq 6). \tag{B}$$

> **Ask yourself** Both conditions involve a choice of fields. How do we combine them without missing a case?

Split on the second choice in (B).

- If $b_3 \geq 6$, then $b_1 + b_2 \leq 3$, so neither $b_1 \geq 6$ nor $b_2 \geq 4$ can hold, which contradicts (A).
- If $b_1 \geq 4$, then together with $b_2 \geq 2$ from (B) and $b_3 \geq 2$ from (A), adding the alternative in (A) gives either

  $$b_1 + b_2 + b_3 \geq 6 + 2 + 2 = 10 \qquad \text{or} \qquad b_1 + b_2 + b_3 \geq 4 + 4 + 2 = 10.$$

  Both are impossible, since the sum is 9.

Every case leads to a contradiction, so no single triple beats both $(5,3,1)$ and $(3,1,5)$. Adam wins with this pair, as required.

Checks

There are only $\binom{11}{2} = 55$ triples, so everything can be checked by brute force. A computer confirms that every one of Adam's 55 triples is beaten by Betty's recipe above, and that every winning reply wins exactly two fields. It also lists the 13 triples that beat $(5,3,1)$ and the 13 that beat $(3,1,5)$, and finds no triple common to the two lists. For a concrete case, $(4,2,3)$ beats $(3,1,5)$ by 2–1, but against $(5,3,1)$ it wins only field 3 and loses fields 1 and 2, so it loses 1–2.

Common slips

- Trying $(3,3,3)$ for Adam's "never win" triple. Betty can play $(2,2,5)$ and lose 1–2, so Adam *can* win with it.
- Forgetting that a win for Betty with only one field won needs Adam to score nothing, and then not using the sum 9 to rule it out.
- In the two-triple part, checking only a few natural replies such as $(6,0,3)$ or $(0,2,7)$ instead of proving that *no* reply works.

Beyond STEP

Colonel Blotto is a classic of game theory. Because Betty moves second with full information she always wins, so the interesting version has both players move at once. Then there is usually no best pure strategy, and the solution is a *mixed strategy*: a probability distribution over triples, as guaranteed by von Neumann's minimax theorem. The two-triple variant in this question is a first step in that direction, since Adam's best defence is to spread his choice over several triples. For problems in the same spirit, see Siklos, *Advanced Problems in Mathematics*. For probability tools that support mixed strategies, see Grimmett and Welsh, *Probability: An Introduction*.

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